

DT124A – Lab 4 report

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Authorship statement:

I, Gagan Adhikari, declare that this lab report is the result of my own original work. Any outside sources used have been properly cited and acknowledged. This report has not been submitted previously for any academic or professional purpose. All program design, implementation, and testing were carried out individually.

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1. Introduction

This lab is to provide me with further practices on algorithm development, implementation, and program testing by solving a more complicated problem. The new objectives of this lab include implementing solutions to sub-problems in the form of small functions, and using arrays and strings. We were asked to Develop a Simple CAI Program (The use of computers in education is referred to as computer-assisted instruction (CAI)).

2. Materials and Methods

Random numbers are generated using the rand() function, seeded once at program start with srand(time(NULL)). All arithmetic questions are constrained to ensure results remain within the range 0–99 and are non-negative.

A user-defined structure, struct Question (Data Structure, n.d.), is used to store test questions, correct answers, and user responses. Arrays of this structure allow storage and later presentation of detailed test results. Flowcharts were used to describe the control flow of the program.

3. Results

The CAI program compiled and executed successfully. The program presents a text-based interface and dynamically generates arithmetic questions based on user selections.

3.1 Task

After program start, the user is prompted to enter their name and is then presented with a main menu. In practice mode, the user may choose addition, subtraction, or mixed questions. Ten questions are generated, and incorrect answers are repeatedly prompted until the correct answer is given, with randomized feedback provided after each attempt. (Array, n.d.)

FUNCTION response_right_answer()
SELECT random message from list of correct responses
DISPLAY message
END FUNCTION
FUNCTION response_wrong_answer()
SELECT random message from list of wrong responses
DISPLAY message
END FUNCTION

In test mode, fifteen questions are generated with no feedback during the test. After
completion, the final score is displayed as a percentage along with a detailed table showing questions, correct answers, and user responses.

```
FUNCTION test()
 DECLARE quiz[15] AS Question
 score ← 0
 FOR i = 0 TO 14
  IF random choice = addition
    THEN add_test(quiz, i)
 ELSE sub_test(quiz, i)
 END
 IF PROMPT "Q (i+1): quiz[i].text = " READ quiz[i].user IF quiz[i].user =
quiz[i].correct
  THEN
  score ← score + 1
END IF
END FOR
  percentage ← (score / 15) * 100
  DISPLAY "Your test result is percentage%"
 DISPLAY "Detailed questions and answers:"
FOR i = 0 TO 14
  DISPLAY i+1, quiz[i].text, quiz[i].correct, quiz[i].user
END FOR
END FUNCTION
```

➔ ***Top-down design***

Main Program

- *Reads user name.*
- *Loops until exit: shows menu, reads choice, calls practice or test.*

- ***Practice***
  - *User chooses addition, subtraction, or mixed.*
  - *Calls the corresponding function (add, sub, mix).*
- ***Test***
  - *Generates 15 questions using add_test or sub_test.*
  - *Collects answers, calculates score, prints summary.*
- ***Feedback***
- *response_right_answer → prints random positive message.*
- *response_wrong_answer → prints random encouragement to retry.*

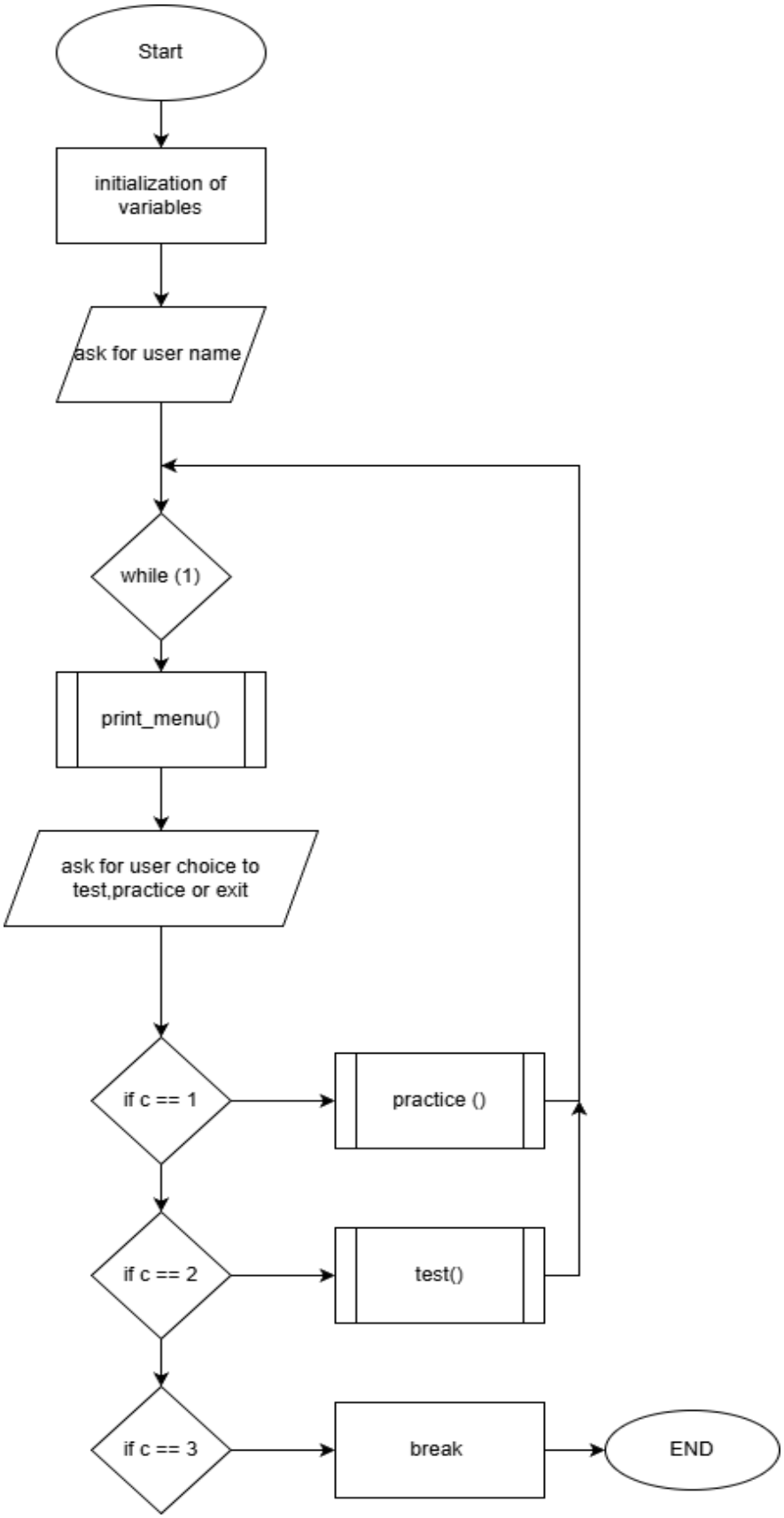


Figure: Flowchart for CAI program.

3.1.1 Discussion

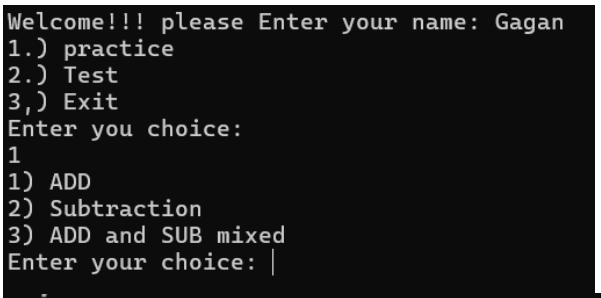
The program demonstrates effective structured programming through the use of small, well-defined functions. Practice mode supports learning by allowing unlimited attempts and immediate feedback, while test mode evaluates performance objectively. The use of structures and arrays improves data organization and enables detailed result reporting.

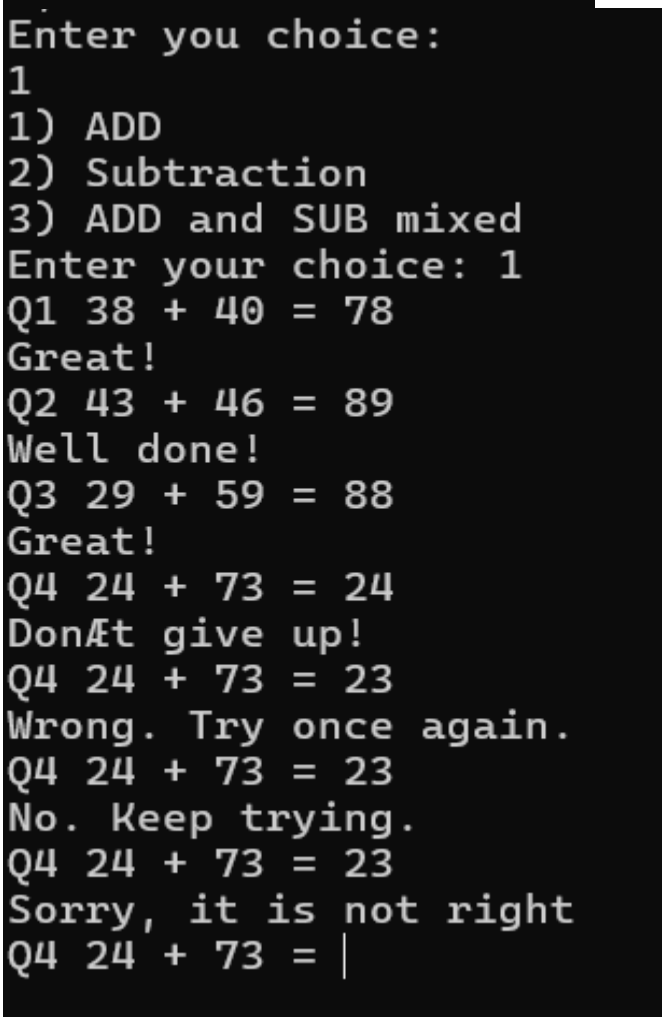
Pseudo Code for main functions only:

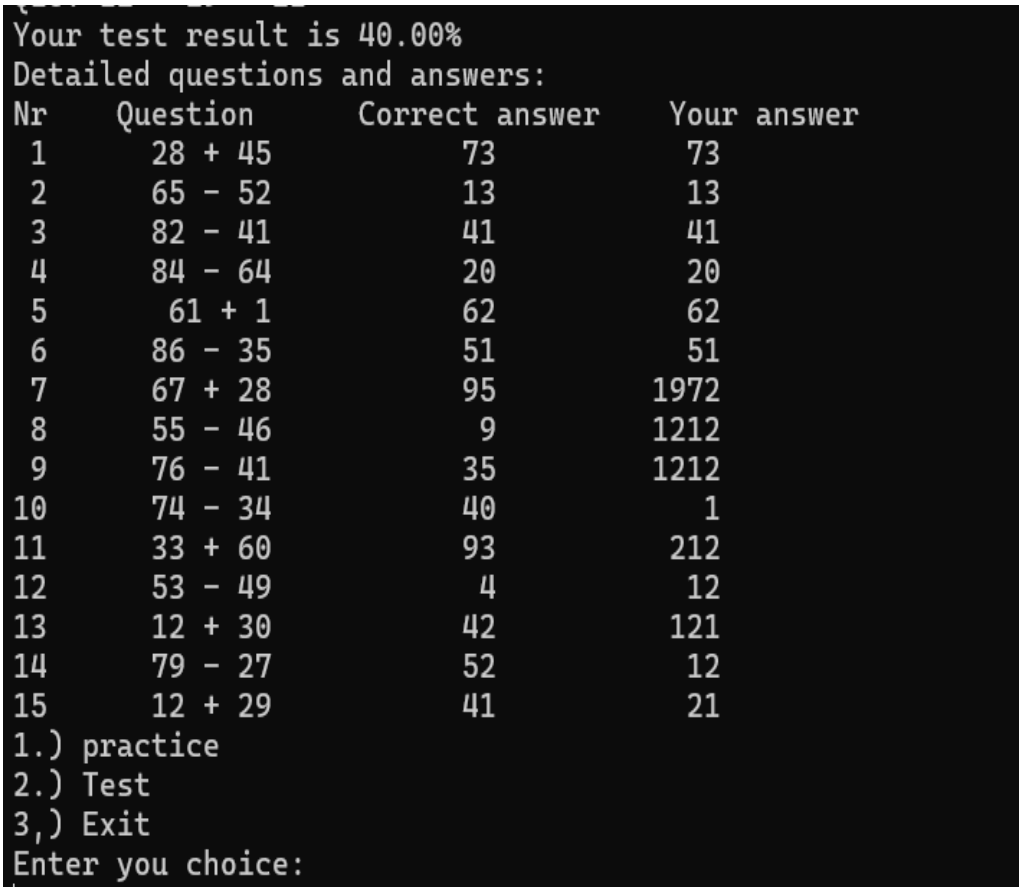
```
MAIN PROGRAM
  PROMPT "Enter your name"
  READ name
  REPEAT
    print_menu()
    PROMPT "Enter your choice"
    READ c
    IF c = 1 THEN
      practice()
    ELSE IF c = 2 THEN
      test()
    ELSE IF c = 3 THEN
      EXIT PROGRAM
    END IF
  UNTIL exit chosen
END PROGRAM
```

4. Test Cases

Here are some test images of the program:


Welcome!!! please Enter your name: Gagan
1.) practice
2.) Test
3,) Exit
Enter you choice:
1
1) ADD
2) Subtraction
3) ADD and SUB mixed
Enter your choice:


Enter you choice:
1
1) ADD
2) Subtraction
3) ADD and SUB mixed
Enter your choice: 1
Q1 38 + 40 = 78
Great!
Q2 43 + 46 = 89
Well done!
Q3 29 + 59 = 88
Great!
Q4 24 + 73 = 24
DonÆt give up!
Q4 24 + 73 = 23
Wrong. Try once again.
Q4 24 + 73 = 23
No. Keep trying.
Q4 24 + 73 = 23
Sorry, it is not right
Q4 24 + 73 =



5. References

Array. (n.d.). Retrieved from greekforgreek: https://www.geeksforgeeks.org/quizzes/arrays-in-c-1/

Data Structure. (n.d.). Retrieved from greekforgreek: https://www.geeksforgeeks.org/c/structures-c/

6. *Appendix:*

→Code has been provided with the report.

→Main Program

print_menu

→ practice

add

sub

mix

→test

add_test

sub_test

→feedback

response_right_answer

response_wrong_answer